

Curated Research Articles

Generated: 2026-08-10 03:51

- **Breaking the Polarity–Coordination Coupling in Electrolyte Design for High-Voltage Lithium Batteries** — score: 1.000 This article presents a novel electrolyte design strategy that separates high solvent polarity from strong lithium ion (Li⁺) coordination, utilizing 2,2,2-trifluoro-N,N-dimethylethylamide (DMTFA) to create a pseudo-weakly solvating electrolyte. This innovation lowers the barriers for Li⁺ desolvation, resulting in enhanced ionic conductivity and stability, facilitating durable, high-voltage cycling and fast-charging capabilities in lithium batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **High-Entropy-Tailored d–p Orbital Hybridization in Na₃V₂(PO₄)₂O₂F Cathodes for Durable Sodium-Ion Batteries** — score: 1.000 The article presents a novel high-entropy cathode material, Na₃V₂(PO₄)₂O₂F, for sodium-ion batteries that enhances charge distribution through modified d–p orbital hybridization. By incorporating multivalent cations, this design achieves a significant reversible capacity of 127.2 mAh g^{−1} and maintains 81.1% capacity retention after 2500 cycles, addressing issues of poor conductivity and structural stability in traditional polyanionic compounds. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Networked Solid Polymer Electrolyte Enabling 5C Fast Cycling and Enhanced Stability of Oriented LiCoO₂ All-Solid-State Thin Film Batteries** — score: 1.000 The article presents a novel networked solid polymer electrolyte (NSPE) that significantly enhances the performance of oriented LiCoO₂ (LCO) thin film batteries. This new electrolyte design allows for stable operation at high voltages (up to 4.5 V) and fast cycling rates of 5C, achieving over 500 cycles without significant degradation, thereby surpassing traditional liquid electrolyte systems in both cycling stability and high-voltage performance. Additionally, the formation of a LiF-rich cathode-electrolyte interphase (CEI) is indicated as a key factor in improving lithium ion transport and mitigating phase transitions during high-stress operation. Journal: *Wiley: Small Methods: Table of Contents*
- **Synergistic Considerations of Deposition–Stripping Mechanisms for Designing Highly Reversible Zinc Metal Anodes** — score: 1.000 The article reviews the intricacies of zinc deposition and stripping processes relevant to improving the performance of aqueous zinc-ion batteries (AZIBs). It emphasizes the coupled nature of these processes and explores various factors influencing zinc behavior, including microstructural and electrochemical aspects. The review calls for further research on stripping dynamics and interfacial stability to enhance the design of reversible zinc metal anodes. Journal: *Wiley: Small Methods: Table of Contents*
- **Fluoride-Ion Batteries: Unraveling the Fundamental Science for Next-Generation Energy Storage** — score: 1.000 This review article explores the current challenges and advances in the development of fluoride-ion batteries (FIBs), highlighting the need to address intrinsic conflicts in electrode and electrolyte performance. It discusses specific issues such as volume expansion in conversion-type electrodes and ionic conductivity in solid electrolytes, emphasizing the importance of advanced characterization methods and interface modulation strategies to enhance battery reliability and efficiency. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Direct Optical Visualization of Kinetic Homogenization in Al-Doped LiCoO₂ Cathodes** — score: 1.000 The study employs optical scattering microscopy to investigate phase transitions in single LiCoO₂ (LCO) particles during high-voltage cycling, revealing that undoped LCO experiences kinetic degradation through asymmetric phase boundaries. In contrast, Al-doped LCO demonstrates improved stability by maintaining a homogenous lithium distribution and suppressing phase separation, attributed to a thermodynamically smoothed free-energy landscape, which ultimately enhances the cycling performance of the cathode materials. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Magnetic Insights Into Heterogeneous and Nonequilibrium Electrochemical Reactions** — score: 1.000 The article presents magnetic characterization as an innovative technique for investigating heterogeneous and nonequilibrium electrochemical reactions, which conventional methods struggle to

analyze. By employing operando monitoring of magnetic properties, researchers can uncover critical electronic state changes and intermediary processes in energy systems, paving the way for advancements in battery and electrocatalyst design. The perspective also discusses the potential for future developments in this field, emphasizing the integration of magnetic techniques with other analytical methodologies. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Halide Solid-State Electrolytes for all-Solid-State Lithium Batteries: Structure, Transport, Interface Relationships and Design Principles** — score: 1.000 The review article discusses advancements in halide solid-state electrolytes (HSEs) for lithium batteries, focusing on the interrelationship between their crystal and defect chemistry, and the resulting ionic conduction properties. It emphasizes the importance of structural disorder and cation substitution for improving conductivity and evaluates the compatibility of HSEs with lithium metal and high-voltage cathodes, while proposing design principles for future scalable and stable applications in solid-state batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Reconstructing Solvation Structure via Strongly Nucleophilic Anions for High-Rate and Low-Temperature Graphite||LFP Pouch Cells in PC-Based Electrolyte** — score: 1.000 The article presents a novel strategy using nucleophilic anions, specifically nitrate (NO_3^-), to enhance the solvation structure of lithium ions in propylene carbonate (PC)-based electrolytes. This approach improves ion transport and solid-electrolyte interphase formation, resulting in significantly better performance of lithium-graphite and lithium iron phosphate pouch cells under high-rate and low-temperature conditions, enabling longer cycling stability and higher capacity than traditional electrolytes. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Perfluorinated Asymmetric Magnesium Salts Enable Stable and Dendrite-Free Magnesium Batteries** — score: 0.900 The study introduces a perfluorinated asymmetric magnesium salt ($\text{Mg}(\text{PFBS})_2$) that significantly improves the stability and performance of rechargeable magnesium batteries by regulating ion deposition and mitigating dendrite formation. By optimizing the electric double layer and forming a dense solid electrolyte interphase (SEI), the cells exhibit exceptional cycling stability for over 1800 hours, high Coulombic efficiency, and sustained performance in full cells with high-rate capabilities. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **In Situ Interfacial Reconstruction Enables Sodiophilic and Ion-Conductive Current Collectors for Anode-Free Sodium Metal Batteries** — score: 0.900 This study presents a novel in situ interfacial reconstruction technique that enhances the performance of anode-free sodium metal batteries by creating a NaF-rich composite interphase on an aluminum current collector. The developed interface, which incorporates metallic Sn and AlF_3 , significantly improves sodium nucleation, ion transport, and mechanical stability, leading to exceptional Coulombic efficiency and sustained capacity in battery cycling, making it a promising solution for high-energy-density storage applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Biomimetic Multilayer Solid-Electrolyte Interphase Customization for Long-Lifespan Practical Aqueous Zinc-Metal Batteries** — score: 0.900 The article presents a novel gradient biomimetic multilayer solid-electrolyte interphase (SEI) for aqueous zinc-metal batteries, drawing inspiration from plant epidermis. This SEI design effectively mitigates dendrite formation and side reactions through a hydrophobic outer layer, a stabilizing middle layer, and a guiding inner layer, resulting in over 2250 hours of cycling life and enhanced performance at high current densities, demonstrating potential for practical applications in aqueous batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Achieving Synergistic Kinetics and Stability in Li-Rich Mn-Based Oxides via AlPO_4 -Assisted Multi-Scale Regulation** — score: 0.800 The study introduces an innovative AlPO_4 -assisted lithiation method to enhance the performance of Li-rich Mn-based oxides by optimizing morphology, stabilizing interfaces, and doping the bulk structure. This approach results in impressive electrochemical performance, achieving around 240 mAh/g with 82.5% retention over 500 cycles at room temperature, and demonstrating strong high-temperature stability, ensuring its practical applicability in high-energy batteries. Journal: *Wiley: Advanced Energy Materials: Table of*

- **Garnet-Induced Amorphization of Fluoride Solid Electrolyte Interlayer Enables High-Voltage and Large-Capacity Lithium Metal Solid-State Batteries** — score: 0.800
The study introduces a novel approach to enhance the performance of lithium metal solid-state batteries by utilizing a garnet-induced amorphization of a fluoride solid electrolyte, Li₂SiF₆, which significantly improves ionic conductivity and reduces interfacial resistance. This advancement allows for high critical current densities and stable cycling performance at elevated voltages, culminating in impressive reversible capacities for the all-solid-state conversion batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Mechanism-Driven Atomic-Level Engineering Over RuO₂-Based Catalysts for Proton Exchange Membrane Water Electrolysis** — score: 0.800
This review article explores recent advancements in RuO₂-based nanocatalysts used for the oxygen evolution reaction (OER) in proton exchange membrane water electrolysis (PEMWE), addressing their stability and kinetic limitations. It distinguishes itself by focusing on atomic-level engineering strategies to improve both catalytic activity and stability, while also emphasizing the need for a deeper understanding of OER mechanisms through advanced characterization techniques. The review concludes by identifying key challenges and future research directions needed to enhance the practical application of these catalysts. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Multidimensional Nanostructure Engineering in Practical Lithium-sulfur Batteries** — score: 0.800
This review explores innovative approaches in cathode material design for lithium-sulfur batteries, emphasizing the role of multidimensional nanostructure engineering. It classifies advancements into strategies that enhance performance through various dimensional synergies, addressing key challenges such as slow kinetics and polysulfide shuttling, and discusses future prospects for improved battery performance. Journal: *Wiley: Small Methods: Table of Contents*
- **Peukert-Informed Prediction and Screening of Manganese-Based High-Rate Potassium Cathode** — score: 0.800
The article presents a novel machine-learning framework that integrates Peukert's law with Random Forest algorithms to enhance the prediction and screening of manganese-based cathodes for potassium-ion batteries. By analyzing over 14,000 compositions, the framework effectively identifies optimal dopant combinations that improve rate performance while preserving capacity, demonstrating a significant advancement in the efficient discovery of high-rate energy storage materials. Journal: *Wiley: Small Methods: Table of Contents*
- **Interfacial Nucleation Processes With SEI Governed for Zinc Ion Batteries** — score: 0.800
This article discusses the development of an organic-inorganic hybrid solid-electrolyte interphase (SEI) that effectively regulates zinc nucleation processes in aqueous zinc-ion batteries. By promoting preferential growth along the (101) crystal plane, the engineered SEI significantly reduces dendrite formation, leading to highly reversible zinc plating/stripping and achieving over 6000 hours of stable cycling, thereby offering insights into enhancing battery performance. Journal: *Wiley: Small Methods: Table of Contents*
- **Dynamic Li⁺ Releasing-Pulling Design of Weakly Coordinated Gel Polymer Electrolytes Realizing Practical Wide-Temperature Lithium Metal Batteries** — score: 0.800
The study presents a novel gel polymer electrolyte (NSP-GPE) designed with a dynamic “releasing-pulling” mechanism to enhance Li⁺ transport for lithium metal batteries, enabling them to function efficiently across a wide temperature range of -20 to 60°C. The electrolyte achieves this by utilizing weak interactions between Li⁺ and the polymer as well as the solvent propylene carbonate, leading to improved safety, faster ion conduction, and high energy and power densities in practical applications, such as powering UAVs. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Progress and Perspectives of In Situ Polymerized Solid-State Electrolytes for Sodium Batteries** — score: 0.800
The article reviews the use of in situ polymerization for creating solid-state electrolytes in sodium batteries, highlighting its advantages over conventional liquid electrolytes such as improved safety and electrode contact. It assesses recent advancements in synthesis methods,

electrolyte compositions, and performance metrics, while also identifying ongoing challenges and outlining future research directions to enhance the technology’s viability for large-scale energy storage applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Regulating the Spin State of Fe-N4 Sites via Electronic Metal-Support Interactions Enables Chloride-Tolerant Bifunctional Oxygen Catalysis** — score: 0.800 The study presents a novel approach to enhance chloride tolerance in oxygen catalysis for seawater-based zinc-air batteries by utilizing electronic metal-support interactions (EMSIs) to regulate the spin state of Fe-N4 sites. By combining these sites with low-content PtFeCu alloy nanoparticles, the transition from low-spin to intermediate-spin configurations improves electrocatalytic activity, yielding significant performance metrics suitable for practical energy storage applications in chloride-rich environments. Journal: *Wiley: Small Methods: Table of Contents*
- **Zinc Reverse Diffusion-Induced Uniform Lithium Deposition in Composite Metal Anodes for High-Performance Lithium Metal Pouch Batteries** — score: 0.800 The article presents a novel in situ metallurgical approach to create a Li/LiZn-Li2O composite anode for lithium-metal batteries, enhancing lithium deposition by using a continuous LiZn network and dispersed Li2O sites. This design enables stable, dense lithium growth and significantly improves cycling stability, achieving impressive performance metrics in pouch cells with high-energy capacity. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Molecular engineering of a nitrate ester additive for ultrastable Ni-rich NCM811 cathodes via anion-rich solvation-structure** — score: 0.800 The article discusses advancements in molecular engineering to enhance the stability of Ni-rich NCM811 cathodes in batteries by utilizing a nitrate ester additive. The study focuses on the development of an anion-rich solvation structure, which plays a crucial role in improving battery performance and longevity. Journal: *ScienceDirect Publication: Nano Energy*
- **Cr/Br Doping Induced Electron Delocalization Tuning D-Band Center Upshift in Na4Fe3(PO4)2P2O7 Cathode for Unlocking Electron/Ion Transport** — score: 0.800 The study explores how Cr and Br doping in Na4Fe3(PO4)2P2O7 cathodes enhances electron delocalization by modifying the electronic configuration and bonding characteristics, thereby improving electron transfer and sodium ion diffusion. This results in a cathode with high capacity and cycling stability, offering a novel approach to optimize polyanionic materials for sodium-ion batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Facilitated Phase Transformation for Improved Performance of Earth-Abundant Disordered Rocksalt Materials** — score: 0.800 The article discusses a novel approach to improve the performance of manganese-rich disordered rocksalt (DRX) materials for lithium-ion batteries by facilitating their transformation into a partial spinel-like -phase through electrochemical delithiation and heating. This method enhances capacity and rate capability while avoiding lengthy cycling processes, establishing a relationship between transformation kinetics and the synthesis parameters. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Defect-driven electronic reconstruction enables closed-pore engineering in hard carbon for superior sodium storage** — score: 0.800 The article discusses a novel approach to enhancing sodium storage capacity in hard carbon materials by utilizing defect-driven electronic reconstruction to engineer closed pores. This method demonstrates a significant improvement in storage performance, showcasing its potential for advancing energy storage technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Regulating Uniform Zinc Deposition via Covalently Tethered Imidazolium Cations in Polyacrylamide Hydrogel Electrolytes for Stable Aqueous Zinc Batteries** — score: 0.800 The article presents a novel hydrogel electrolyte, incorporating covalently tethered imidazolium cations, designed to enhance the stability of aqueous zinc-ion batteries by mitigating issues like dendrite growth and parasitic reactions. This electrolyte forms a protective electrostatic layer at the zinc anode, enabling uniform zinc deposition and significantly extending battery lifespans, as demonstrated by impressive

cycle stability in both symmetric and full cell configurations. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Resolving the Mechano-Electrochemical Dilemma in Structural Batteries via In Situ Entropy Engineering** — score: 0.800 The article introduces an innovative entropy engineering approach to enhance the performance of structural batteries (SBs) by creating a robust nickel-rich cathode that mitigates capacity degradation and mechanical failure from electrochemical and mechanical stresses. Through multi-element co-doping, the cathode demonstrates impressive metrics, including a specific capacity of 122.34 mAh g⁻¹ with 82.25% retention over 200 cycles, alongside strong mechanical properties, thereby addressing the challenges of combining high energy density with structural integrity for massless energy storage applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **3D Micro-Network Li/Al₄Li₉ Composite Anodes Constructed by One-Step Melt-Spinning for High Energy Density Lithium Metal Batteries** — score: 0.700 The study presents a novel 3D micro-network Li/Al₄Li₉ composite anode fabricated through one-step melt-spinning, which enhances lithium plating and reduces structural degradation during battery cycling. This innovative anode architecture ensures stable cycling performance, showcasing its potential for high-energy-density lithium metal batteries with significantly improved cycling stability and resistance to volume changes. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Bridging Materials and Performance in Batteries: The Role of Accurate Models in Electrode Design and Manufacturing** — score: 0.600 The article discusses how the design of porous battery electrodes significantly impacts performance, highlighting the challenges posed by ion and electron transport limitations. It reviews various mathematical modeling approaches, from semi-empirical to physics-based, which help identify critical design parameters and improve manufacturing processes. The focus is on optimizing electrode architecture to enhance performance while addressing the transition from model insights to scalable production. Journal: *Wiley: Small Methods: Table of Contents*
- **Nano-Heterostructured TiO₂-TiO-TiN/MoO₃-x-Mo₂N Composite LIB Anode With Enhanced Li⁺ Storage via Glycine-assisted Anodization and Cathodic Deposition** — score: 0.600 This article presents a novel, scalable method for fabricating a high-performance lithium-ion battery anode using a nano-heterostructured TiO₂-TiO-TiN/MoO₃-x-Mo₂N composite on titanium substrates. The anode achieves a remarkable areal capacity of 2639 μ Ah cm⁻² and significantly lowers charge-transfer resistance, demonstrating the effectiveness of integrating glycine-assisted anodization with cathodic deposition to enhance lithium-ion storage capabilities and promote efficient energy delivery. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Computational Discovery of Novel Sulfide Lithium Superionic Conductors via a Structural-Feature-Constrained Generative Model** — score: 0.600 This study presents a generative modeling approach that utilizes structural feature constraints to effectively discover new superionic conductors, specifically lithium sulfides. By focusing on a favorable body-centered cubic anion framework, the researchers generated 21 novel ternary sulfides that display promising ionic conductivities, thereby demonstrating the model's effectiveness and providing a broader strategy for material design based on structural properties rather than direct property constraints. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Polymer-induced interfacial nucleation of a soft-interior/rigid-exterior ZIF-71@PMN dual-layer interphase for highly stable Zn anodes** — score: 0.400 The article presents the development of a dual-layer interphase, ZIF-71@PMN, which features a soft interior and a rigid exterior, aimed at enhancing the stability of zinc anodes. This innovative approach utilizes polymer-induced interfacial nucleation to improve performance, potentially addressing the challenges in zinc-based battery technology. Journal: *ScienceDirect Publication: Nano Energy*
- **A Sol-Gel Derived Sodium Iron Pyrophosphate Sulfate Cathode Material for High-Performance Sodium-Ion Batteries** — score: 0.400 The article presents a novel sodium iron pyrophosphate sulfate (NFPS) cathode material developed using a sol-gel method and biomass-derived carbon, achieving an initial capacity of 101 mAh g⁻¹ and high cycling stability at varying rates and low

temperatures. The findings suggest this environmentally friendly material could significantly enhance the performance and scalability of sodium-ion batteries for large-scale energy storage applications. Journal: *Wiley: Small Methods: Table of Contents*

- **Trace Copper-Mediated Crystal Structure Modulation Enables Superior Stable Sodium Storage in NASICON Cathodes** — score: 0.400 The study explores how trace amounts of copper can effectively alter the crystal structure of NASICON (sodium superionic conductor) cathodes, leading to enhanced stability and performance in sodium storage applications. This modulation of crystal structure significantly improves the longevity and efficiency of sodium-ion batteries, positioning them as a viable alternative for energy storage solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Anion-immobilizing poly(ionic liquid)-based semi-interpenetrating electrolyte enabling ultrafast Li⁺ transport in solid-state lithium metal batteries** — score: 0.400 The article discusses the development of a novel semi-interpenetrating polymer electrolyte that incorporates anion-immobilizing poly(ionic liquid) to facilitate rapid lithium ion transport in solid-state lithium metal batteries. This innovative approach aims to enhance battery performance by improving ionic conductivity, which is crucial for next-generation energy storage systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **In situ pore sealing of cellulose derived hard carbon via coal tar cross linking for high capacity sodium ion batteries** — score: 0.400 The article discusses a novel approach for enhancing sodium-ion batteries by in situ sealing the pores of cellulose-derived hard carbon using coal tar cross-linking. This method aims to improve the material's capacity and stability, potentially leading to more efficient and durable battery systems. The findings are set to be published in October 2026 in Nano Energy. Journal: *ScienceDirect Publication: Nano Energy*
- **Noncovalent Interaction-Regulated Polymerizable Deep Eutectic Solvent Electrolyte for Sodium Metal Batteries** — score: 0.200 The article discusses the development of a novel polymerizable deep eutectic solvent electrolyte that utilizes noncovalent interactions to enhance the performance of sodium metal batteries. This advanced electrolyte aims to improve battery efficiency and stability, contributing to the progression of energy storage technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Constructing 3D Ion-Conductive Binder Networks for Boosted Prelithiation Kinetics and Interfacial Stability of High-Loading Si/C Anodes** — score: 0.200 The article discusses the development of 3D ion-conductive binder networks that enhance the kinetics of prelithiation and improve interfacial stability in high-loading silicon/carbon anodes. This innovation aims to address the challenges associated with lithium-ion batteries, particularly in terms of performance and longevity. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Advanced Manufacturing Routes for Electrodes and Cells in All-Solid-State Batteries: From Powder to Power** — score: 0.200 The article reviews advanced manufacturing techniques for electrodes and cells in all-solid-state batteries (ASSBs), emphasizing the importance of both wet and dry processing methods. It explores the challenges of solvent compatibility and binder design in wet processes, alongside the benefits of dry processing for sustainability and efficiency. Additionally, the review highlights the role of pressure control and bipolar stacking in improving the performance and density of ASSBs, ultimately aiming to bridge the gap between laboratory innovation and commercial viability. Journal: *Wiley: Small Methods: Table of Contents*
- **Single-walled carbon nanotube film based self-adhesive air electrode featuring interaction-enhanced dual sites for high-performance Zn-Air batteries** — score: 0.200 The article discusses a novel air electrode for zinc-air batteries, constructed from a self-adhesive film of single-walled carbon nanotubes. This design enhances battery performance by optimizing dual interaction sites, offering potential improvements in energy storage efficiency and overall battery functionality. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Bionic Topology Reconstruction Drives Confined Competitive Ligand-Exchange Hopping**

- Dynamics in Hydrogel Electrolytes to Achieve High-Performance Structural Zn-Ion Batteries** — score: 0.200 This article explores the innovative use of bionic topology reconstruction in hydrogel electrolytes to enhance competitive ligand-exchange hopping dynamics, which significantly improves the performance of structural zinc-ion batteries. The study presents findings that could lead to advancements in energy storage technology by optimizing the ionic conductivity and efficiency of battery systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Microstructure–Electrolyte Compatibility Engineering in Lignite-Derived Hard Carbon for High-Rate Sodium-Ion Batteries** — score: 0.200 The article discusses advancements in the microstructure and electrolyte compatibility of lignite-derived hard carbon, aimed at enhancing high-rate performance in sodium-ion batteries. The authors explore engineering strategies to improve battery efficiency and longevity, positioning this research within the broader context of energy storage innovations. Journal: *ScienceDirect Publication: Energy Storage Materials*
 - **One-pot spray pyrolysis synthesis strategies of lithium-rich manganese-based oxide-spinel composite microspheres as advanced cathode materials for lithium-ion batteries** — score: 0.200 This article discusses a novel one-pot spray pyrolysis method for synthesizing lithium-rich manganese-based oxide-spinel composite microspheres, aimed at enhancing the performance of cathode materials in lithium-ion batteries. The research highlights the efficiency and effectiveness of this synthesis strategy in developing advanced materials for energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
 - **Low-Cost Ambient Pressure Drying Approach for Highly Porous Nanomaterial Structures (Small Methods 15/2026)** — score: 0.200 Researchers developed a cost-effective ambient pressure drying technique to replace traditional critical point drying for highly porous nanomaterials. By employing low surface tension solvents, they were able to evaporate these solvents from fragile aeromaterials made from various compounds like graphene and MXene, effectively preserving their microstructure and preventing collapse of the porous framework. Journal: *Wiley: Small Methods: Table of Contents*
 - **Phenothiazine-Based -Extended Covalent Network Cathodes for Lithium Dual-Ion Batteries with High Energy Density and Fast Charging** — score: 0.200 The article discusses the development of phenothiazine-based covalent network cathodes for lithium dual-ion batteries, which exhibit enhanced energy density and rapid charging capabilities. The research presents a novel approach to improving battery performance through innovative material design, leading to potential advancements in energy storage technology. Journal: *ScienceDirect Publication: Energy Storage Materials*
 - **Design strategy for integrated photo-rechargeable batteries with high energy density** — score: 0.100 The article presents a design strategy for developing integrated photo-rechargeable batteries that achieve high energy density. It details innovative approaches aimed at enhancing battery performance by merging photovoltaic technology with energy storage, thereby addressing challenges in sustainable energy solutions. Journal: *ScienceDirect Publication: Energy Storage Materials*
 - **Engineered interfaces in electronic materials for energy-efficient computing** — score: 0.000 The article discusses the critical role of engineered interfaces in electronic materials as devices become smaller, highlighting their impact on energy efficiency, switching processes, and thermal management. It advocates for tailored interface designs to enhance performance in logic, memory, and interconnects while emphasizing the importance of predictive modeling to advance interface engineering. Journal: *Nature Reviews Materials*
 - **Gasdermin D-mediated delivery of caspase inhibitors to suppress pyroptosis** — score: 0.000 This article discusses the role of Gasdermin D in the targeted delivery of caspase inhibitors, which are designed to effectively suppress pyroptosis, a form of programmed cell death associated with inflammation. The findings suggest potential therapeutic applications for managing diseases characterized by excessive pyroptotic cell death. Journal: *Nature*
 - **Author Correction: Maturation and circuit integration of transplanted human cortical organoids** — score: 0.000 This article addresses corrections to previous findings regarding the mat-

uration and integration of transplanted human cortical organoids within host brains. The revisions clarify experimental methodologies and outcomes, contributing to a more accurate understanding of organoid development and their potential applications in neurological research. Journal: *Nature*